

GRADUATION PROJECT

FortiOS 7.6 Administrator | Creativa

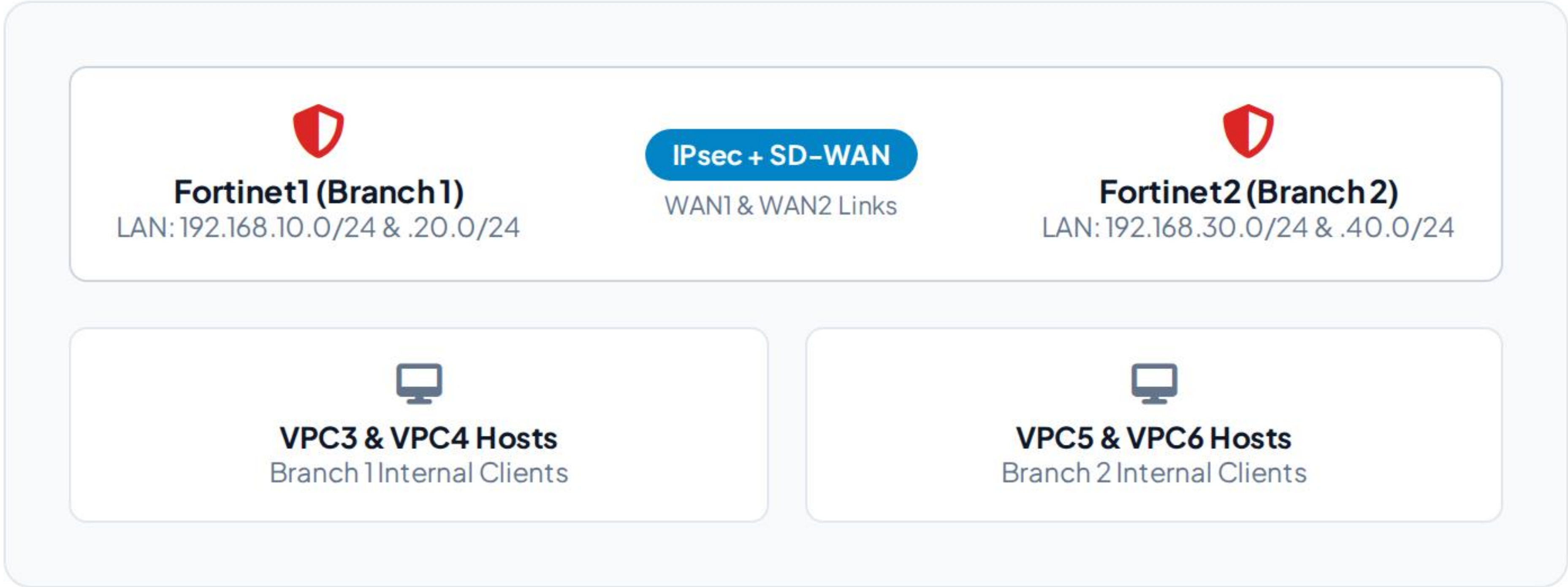
Enterprise Multi-Branch Network Security & SD-WAN

Simulated dual-branch deployment with FortiGate-VM64-KVM, site-to-site IPsec VPN, user authentication, and dynamic WAN redundancy.

 EVE-NG Community Platform

 FortiGate v7.6.7

Simulated Multi-Branch Architecture



Virtualization Platform

EVE-NG Community Edition hosting full virtualized firewall hardware nodes.

Firewall Operating System

FortiOS 7.6.7 (VM64-KVM) enforcing enterprise security policy features.

Dual WAN Topology

Primary WAN1 (10.0.0.0/24) and Secondary WAN2 (20.0.0.0/24) cloud links.

Core Technical Deliverables



Site-to-Site VPN

DES-SHA256 encrypted Phase 1 and Phase 2 IKE tunnels between branch sites.



SD-WAN Zone

Virtual WAN link grouping primary and backup links for continuous uptime.



User Authentication

Local AuthGroup enforcement for outbound web access control on internal hosts.



UTM Security

Deep SSL certificate inspection and Antivirus payload scanning active.

IP Addressing & Interface Mapping

Device	Interface	IP Address / Subnet	Role / Connected Network
Fortinet1	port1 / port4	10.0.0.1 / 20.0.0.1	Primary WAN1 & Secondary SD-WAN WAN2
Fortinet1	port2 / port3	192.168.10.1 / 192.168.20.1	Branch 1 LAN Segment A & Segment B
Fortinet2	port1 / port4	10.0.0.2 / 20.0.0.2	Primary WAN1 & Secondary SD-WAN WAN2
Fortinet2	port2 / port3	192.168.30.1 / 192.168.40.1	Branch 2 LAN Segment A & Segment B
End Hosts	VPC3 - VPC6	192.168.x.10/24	Host Endpoints connected across LAN segments

Static Routing & Firewall Policies

Static Route Strategy

- ✓ **Gateway Mapping:** Directs remote branch LAN subnets through next-hop WAN gateway.
- ✓ **Interface Binding:** Outbound traffic bound explicitly to port 1 WAN interface.
- ✓ **Reciprocal Reachability:** Symmetrical static route rules deployed on Fortinet2.

Firewall Policy Rules

- ✓ **Source NAT:** SNAT enabled on Internal - to - WAN policy for internet access.
- ✓ **Bidirectional Rules:** Pair policies configured to accept return traffic safely.
- ✓ **Service Scoping:** Applied to all standard enterprise services and schedules.

Identity-Based User Authentication



1. Local User Credentials

Created student1 user database with strong password policies.



2. AuthGroup Mapping

Assigned local user accounts to centralized AuthGroup object.



3. Policy Enforcement

Bound AuthGroup to outbound policy requiring authentication for WAN egress.

UTM Security Profiles



Antivirus Inspection

Profile: default

Intercepts malicious file transfers across HTTP, FTP, and SMTP streams in real-time before payloads reach local hosts.



SSL/SSH Certificate Inspection

Profile: certificate-inspection

Validates SSL server certificates, mitigates MITM risks, and provides deep header visibility into encrypted sessions.

Site-to-Site IPsec VPN Tunnel



Phase 1 IKE SA

Configured VPN-F1-F2 interface using DES-SHA256 proposal and PSK key exchange.



Phase 2 IPsec SA

Mapped source 192.168.10.0/24 to destination 192.168.30.0/24 subnets.



Routed VPN Policy

Added firewall policy routing inter-branch traffic directly into the IPsec tunnel.

HIGH AVAILABILITY

SD-WAN Architecture & Dual-WAN

Single WAN Vulnerability

Standard single-ISP configurations lead to complete branch isolation if port1 physical link or ISP upstream service fails.

SD-WAN Zone Solution

Grouped WAN-Link2 (port4) inside virtual-wan-link zone, enabling dynamic path selection and instant failover.

Automatic Failover Operations



Seamless Session Recovery

Traffic re-routed dynamically via virtual-wan-link

Zone Policy Binding

Replaced fixed outgoing interfaces in firewall policy with `virtual-wan-link` zone.

Active Member Validation

Continuous health probes evaluate member gateway metrics to direct outgoing sessions.

TESTING & VALIDATION

CLI Diagnostic Proof

```
# VPN Tunnel Status Verification  
diagnose vpn ike gateway list  
name: VPN-F1-F2  
addr: 10.0.0.1:500 -> 10.0.0.2:500  
status: established (SA 1/1)
```

```
# End-to-End ICMP Ping Test  
VPCS> ping 192.168.30.10  
192.168.30.10 icmp_seq=1 ttl=62 time=1.2ms  
192.168.30.10 icmp_seq=2 ttl=62 time=0.9ms  
Result: 100% Reachability via IPsec
```



Project Complete & Verified

Successfully implemented end-to-end multi-branch network security, IPsec encryption, identity enforcement, and resilient SD-WAN redundancy.

Questions & Discussion